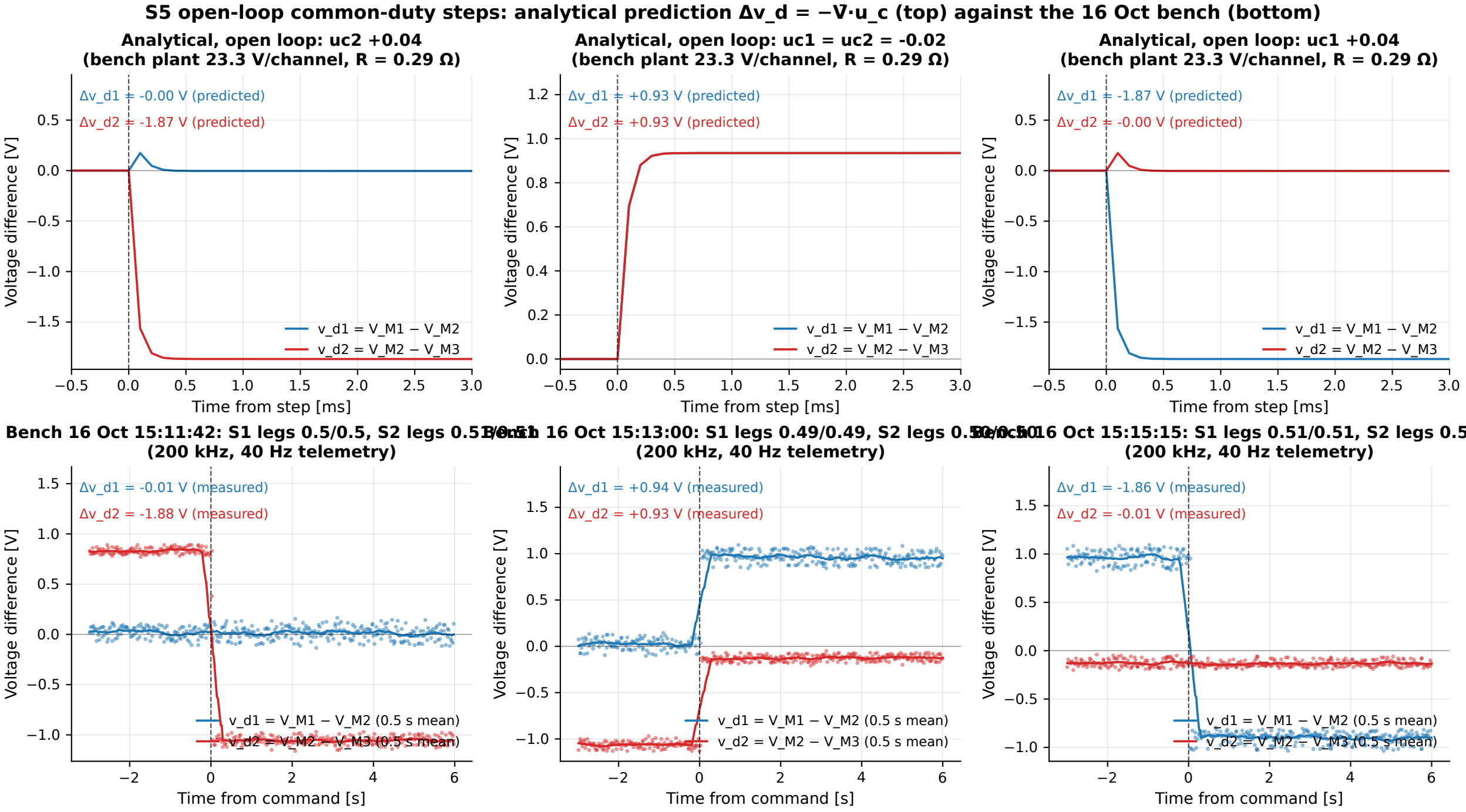

S5 open-loop common-duty steps: analytical prediction Δv_d = −B̄·u_c (top) against the 16 Oct bench (bottom)
Analytical, open loop: uc2 +0.04
(bench plant 23.3 V/channel, R = 0.29 Ω)
Δv_d1 = -0.00 V (predicted)
Δv_d2 = -1.87 V (predicted)
Analytical, open loop: uc1 = uc2 = -0.02
(bench plant 23.3 V/channel, R = 0.29 Ω)
Δv_d1 = +0.93 V (predicted)
Δv_d2 = +0.93 V (predicted)
Analytical, open loop: uc1 +0.04
(bench plant 23.3 V/channel, R = 0.29 Ω)
Δv_d1 = -1.87 V (predicted)
Δv_d2 = -0.00 V (predicted)
Voltage difference [V]
Time from step [ms]
v_d1 = V_M1 − V_M2
v_d2 = V_M2 − V_M3
Bench 16 Oct 15:11:42: S1 legs 0.5/0.5, S2 legs 0.5/0.51
(200 kHz, 40 Hz telemetry)
Bench 16 Oct 15:13:00: S1 legs 0.49/0.49, S2 legs 0.5/0.51
(200 kHz, 40 Hz telemetry)
Bench 16 Oct 15:15:15: S1 legs 0.51/0.51, S2 legs 0.5
(200 kHz, 40 Hz telemetry)
Δv_d1 = -0.01 V (measured)
Δv_d2 = -1.88 V (measured)
Δv_d1 = +0.94 V (measured)
Δv_d2 = +0.93 V (measured)
Δv_d1 = -1.86 V (measured)
Δv_d2 = -0.01 V (measured)
Voltage difference [V]
Time from command [s]
v_d1 = V_M1 − V_M2 (0.5 s mean)
v_d2 = V_M2 − V_M3 (0.5 s mean)